

MTH024 Quiz1

Problem 1

$$A \cup B = \{x | x \in A \text{ or } x \in B\} = \{1, 2, 3, 4, 5, 6, 7\}$$

$$A \cap B = \{x | x \in A \text{ and } x \in B\} = \{4, 5\}$$

Problem 2

Define : E^c is set of odd numbers

$$E^c = \{x \in \mathbb{N} | x \text{ is odd numbers}\} = \{x \in \mathbb{N} | x = 2k + 1, k \in \mathbb{N} \cap \{0\}\}$$

Problem 3

$$x \in A \cap (B \cup C) \Leftrightarrow x \in A \text{ and } x \in B \cup C$$

$$\Leftrightarrow x \in A \text{ and } (x \in B \text{ or } x \in C)$$

$$\Leftrightarrow (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)$$

$$\Leftrightarrow x \in A \cap B \text{ or } x \in A \cap C$$

$$\Leftrightarrow x \in (A \cap B) \cup (A \cap C)$$

Problem 4

$$(a) : \mathbb{Z} \quad \mathbb{R} \quad \mathbb{Q}$$

$$(b) : \mathbb{Q} \quad \mathbb{R}$$

$$(c) : \mathbb{R}$$

$$(d) : \mathbb{Q} \quad \mathbb{R}$$

Problem 5

let a be an even number, $a = 2k$, b is any integer

$$a \times b = 2k \times b = 2(kb)$$

$\therefore 2(kb)$ is even,

$\therefore a \times b$ is even.

Problem 6

Suppose for contradiction that n is an odd number, such that n^2 is even.

$$\therefore n = 2k + 1$$

$$\therefore n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

$$\therefore 2(2k^2 + 2k) \text{ is an even number}$$

$$\therefore 2(2k^2 + 2k) + 1 \text{ is an odd number}$$

But this is not true since n^2 is even.

Hence if n^2 is an even integer, then n must also be an even integer.

Problem 7

Suppose for contradiction that $\sqrt{2}$ is a rational number,

Exist coprime integers p and q ($q \neq 0$) such that $\sqrt{2} = \frac{p}{q}$.

$$\therefore 2 = \frac{p^2}{q^2} \Rightarrow p^2 = 2q^2$$

$$\therefore p \text{ is even, let } p = 2k$$

$$\therefore (2k)^2 = 2q^2 \Leftrightarrow q^2 = 2k^2$$

$$\therefore q \text{ is even.}$$

Since p and q are both even numbers, p and q are not coprime. But it is not true since p

and q should be coprime, then $\sqrt{2}$ is an irrational number.

Problem 8

base: when $n = 1$, $1 = \frac{1 \times (1+1)}{2}$ satisfies the function.

induction: when $n = k$, $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$,

At the same time, when $n = k + 1$, $1 + 2 + \cdots + k + 1 = \frac{k(k+1)}{2} + (k + 1)$

$\therefore \frac{k(k+1)}{2} + (k + 1) = \frac{(k+1)(k+2)}{2}$

$\therefore 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ is proved.

Problem 9

base: when $n = 1$, $1^2 = \frac{1 \times (1+1) \times (2+1)}{6}$ satisfies the function.

induction: when $n = k$, $1^2 + 2^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}$,

At the same time, when $n = k + 1$, $1^2 + 2^2 + \cdots + (k + 1)^2 = \frac{k(k+1)(2k+1)}{6} + (k + 1)^2$

$\therefore \frac{k(k+1)(2k+1)}{6} + (k + 1)^2 = \frac{(k+1)(k+2)(2k+3)}{6} = \frac{(k+1)(k+2)(2(k+1)+1)}{6}$

$\therefore 1^2 + 2^2 + \cdots + n^2 = \frac{k(k+1)(2k+1)}{6}$ is proved.

Problem 10

base: when $n = 4$, $4! = 24 > 2^4 = 16$ satisfies $n! > 2^n$.

induction: when $n = k$, $k! > 2^k$,

At this time, when $n = k + 1$, $(k + 1)! = (k + 1)k! > (k + 1)2^k$

$\therefore n \geq 4$

$\therefore k + 1 \geq 5 > 2$

$\therefore (k + 1)2^k > 2 \times 2^k = 2^{k+1}$

$\therefore (k + 1)! > 2^{k+1}$

$\therefore n! > 2^n$ is proved.